

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/F_DEB_SAALYe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 33113, C2 = 6666, C3 = 2212)

text_priors <- '
Distrib (Fm,          TruncNormal,          0.1,      0.2, 0.01,      0.8);
Distrib (pAm,         TruncNormal_cv,        100,      0.2, 40,      2000);
Distrib (r_pAm_pM,    TruncNormal_cv,        0.8,      0.1, 0.1,      2);
Distrib (v,           TruncNormal_cv,        0.18505,   0.1, 0.001,    2);
Distrib (kap,         Uniform,                0,        1);
Distrib (kapA,        Uniform,                0.1,      0.9);
Distrib (Ehp,         TruncNormal_cv,        100,      0.2, 40,      300);
Distrib (E_coc,       TruncNormal_cv,        30,      0.2, 1,      300);
Distrib (rOM_ClxHorse, LogUniform,          0.00001,   0.3);
Distrib (dv,          TruncNormal_cv,        2,      0.5, 0,      10);
Distrib (wE,          LogUniform,          0.000000001, 0.1);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter)

# ./mcsim.DEBcali DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	Fm.1.	pAm.1.	r_pAm_pM.1.	v.1.	kap.1.	kapA.1.
Result_mode	0.72218500	204.26300	0.74487400	0.18623500	0.93345600	0.68673400

2.5%	0.51074600	177.58895	0.61476800	0.16582100	0.90661470	0.58313900
97.5%	0.79432960	216.11700	0.83713780	0.23028165	0.95911285	0.75227400
Min	0.32098100	90.58550	0.55655200	0.14276100	0.43482600	0.19504900
Max	0.79999700	233.26000	0.92895100	0.25558900	0.96772200	0.81914600
Result_mean	0.68774847	196.02106	0.72347807	0.19559403	0.93386496	0.66420673
Result_sd	0.07575171	10.00471	0.05719509	0.01652855	0.01471575	0.04376183
	Ehp.1.	E_coc.1.	rOM_ClxHorse.1.	dv.1.		wE.1.
Result_mode	105.7100	22.700700	0.002488430	3.0899700	8.414910e-06	
2.5%	68.3435	18.676600	0.001350550	2.3511700	5.812825e-07	
97.5%	137.7530	31.207400	0.004116351	3.7747610	1.097180e-05	
Min	56.5758	13.897500	0.000125481	0.5220400	1.767190e-09	
Max	163.0110	38.358000	0.187905000	4.2290100	1.200400e-05	
Result_mean	101.5760	24.719303	0.002530767	3.0722812	8.359512e-06	
Result_sd	18.5861	3.212154	0.001286483	0.3690796	2.004390e-06	
	Sigma_W.1.					
Result_mode	0.11579500					
2.5%	0.10352710					
97.5%	0.13210795					
Min	0.09382190					
Max	4.53346000					
Result_mean	0.11718770					
Result_sd	0.03972208					

Number of observations, n = 277
 Number of parameters, k = 11
 Log-likelihood, LL = 10.01913
 BIC = 41.82593

Potential scale reduction factors:

	Point est.	Upper C.I.
dv.1.	1.20	1.59
E_coc.1.	1.02	1.07
Ehp.1.	1.37	1.97
Fm.1.	1.03	1.10
kap.1.	1.42	2.10
kapA.1.	1.01	1.01
pAm.1.	1.01	1.04
r_pAm_pM.1.	1.13	1.39
rOM_ClxHorse.1.	1.00	1.00
Sigma_W.1.	1.00	1.00

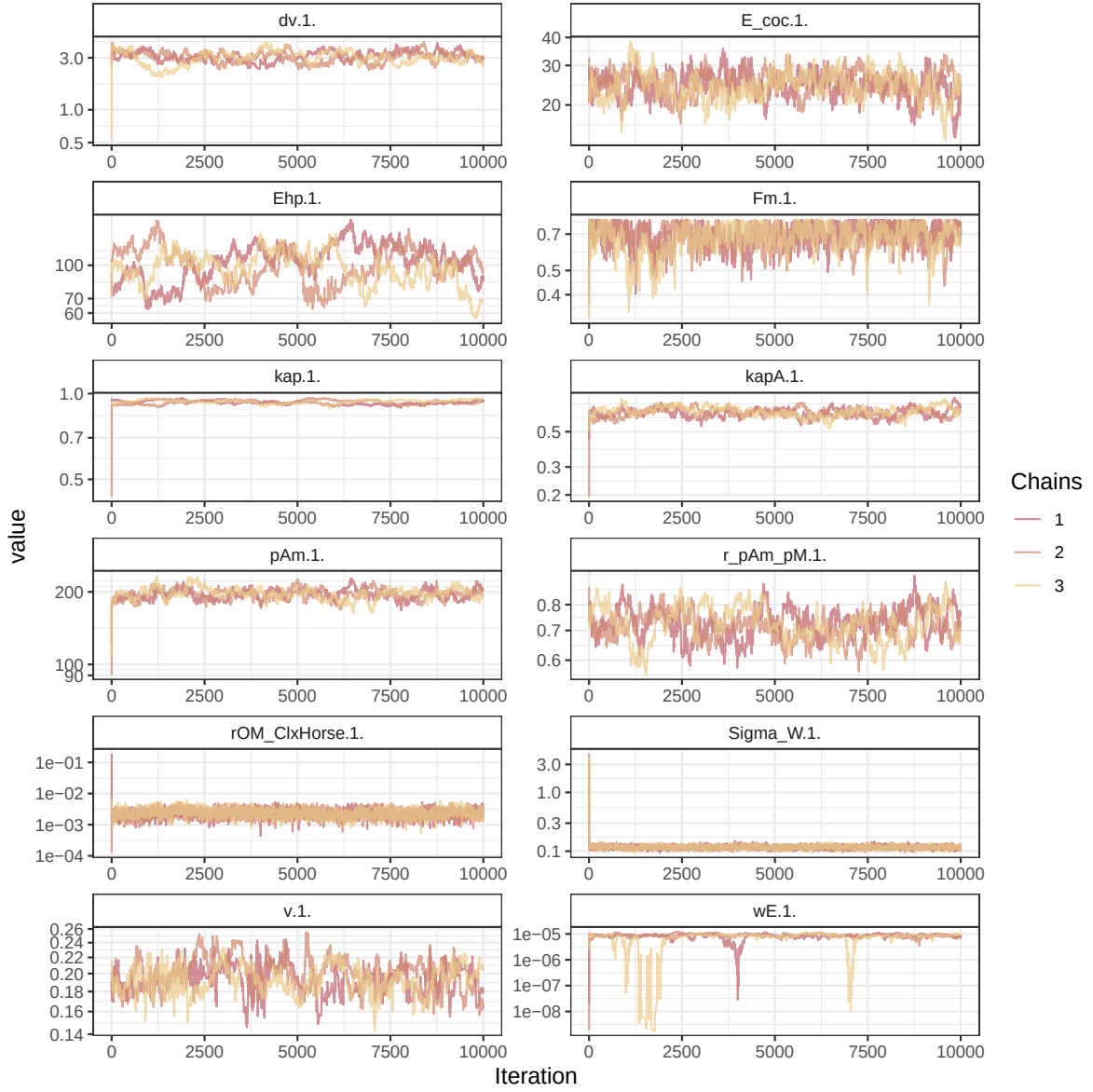


Figure 1: Parameters chains. Only the first iteration and the last r `Nb_iter_kept` are represented.

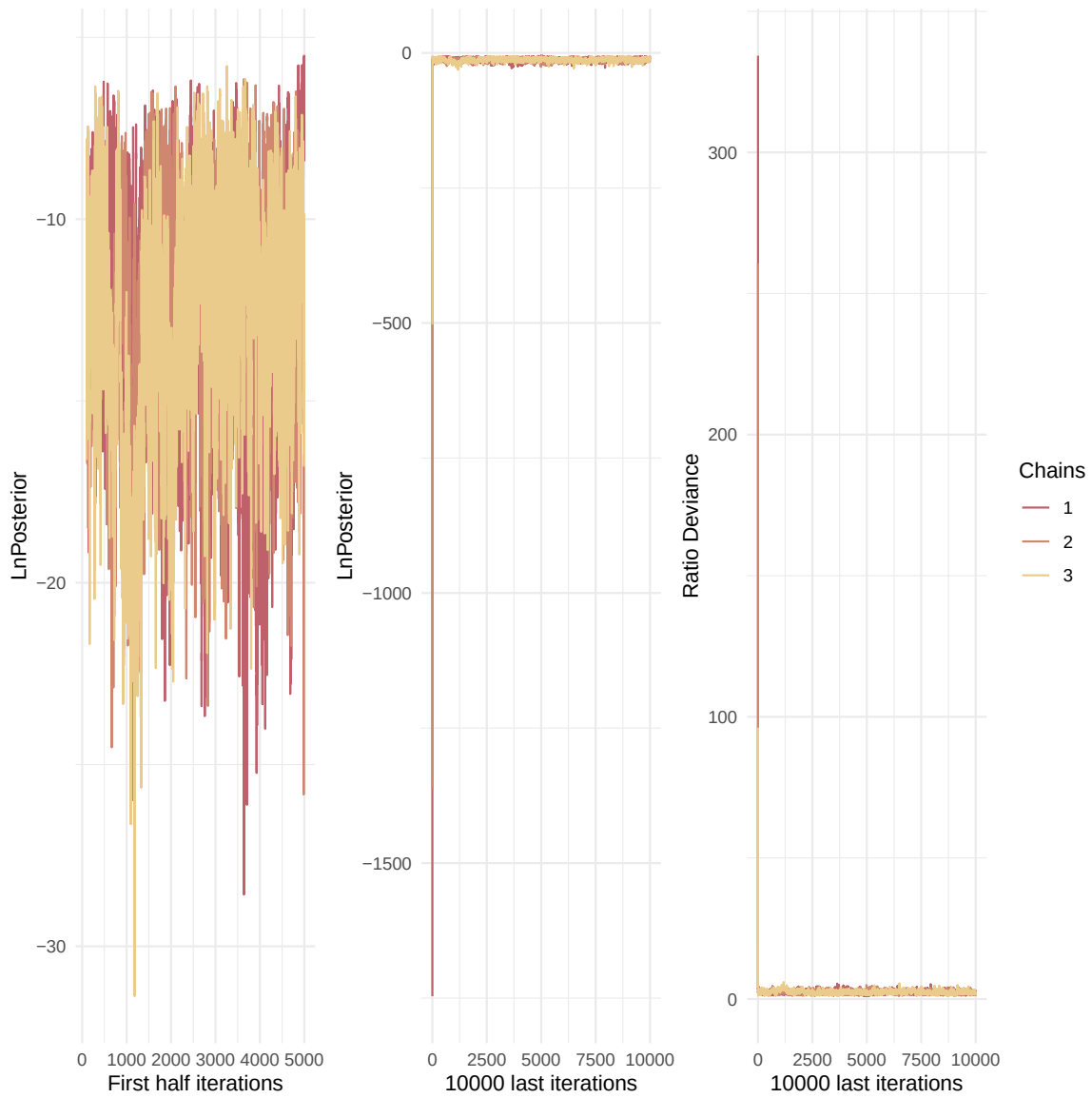


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

v.1.	1.08	1.25
wE.1.	1.10	1.18

Multivariate psrf

1.43

```
mcmc_list_corr <- mcmc.list(  
  lapply(mcmc_list, function(x) {  
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc  
  })  
)  
  
colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))  
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

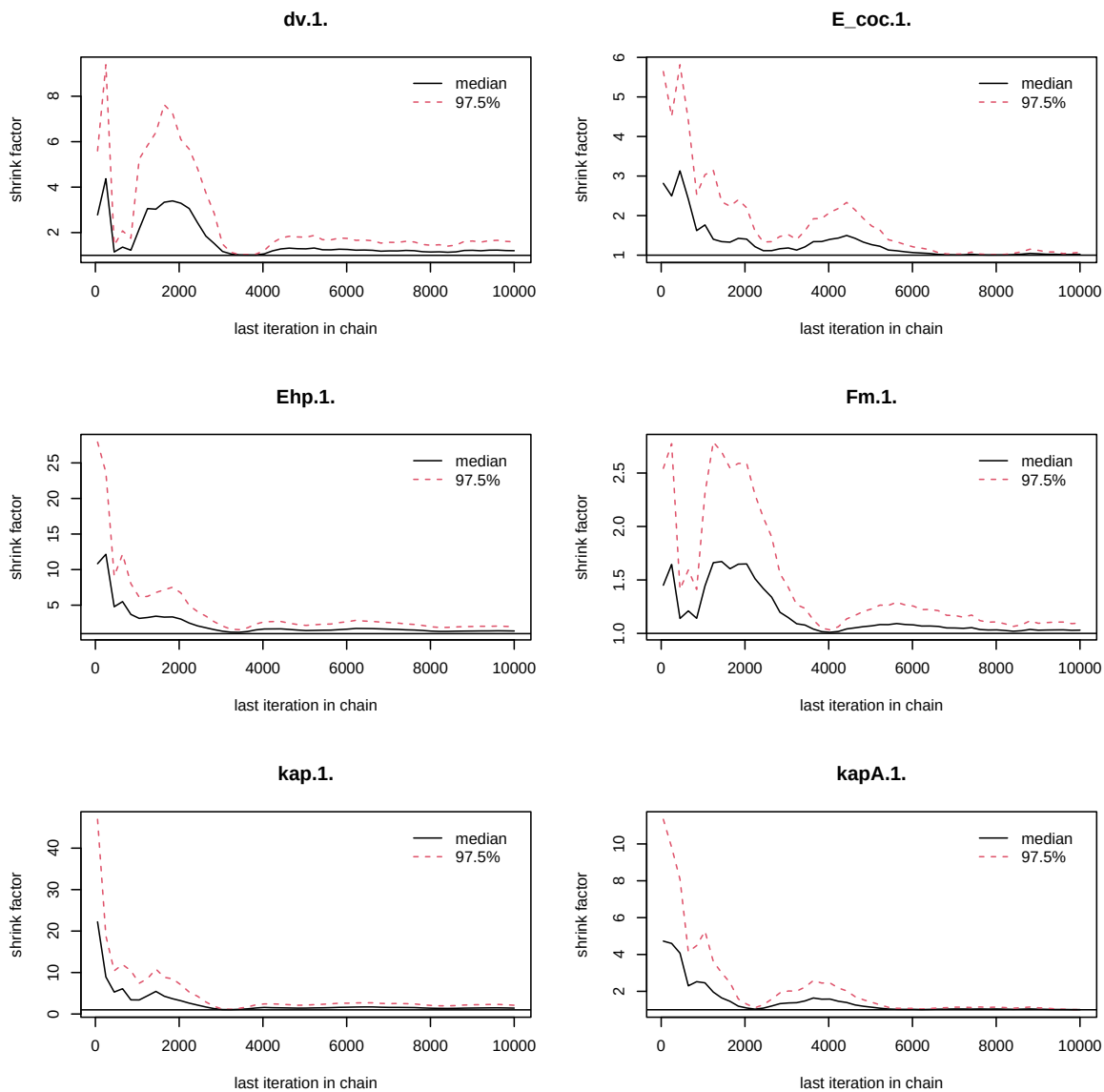


Figure 3: Chains convergence.

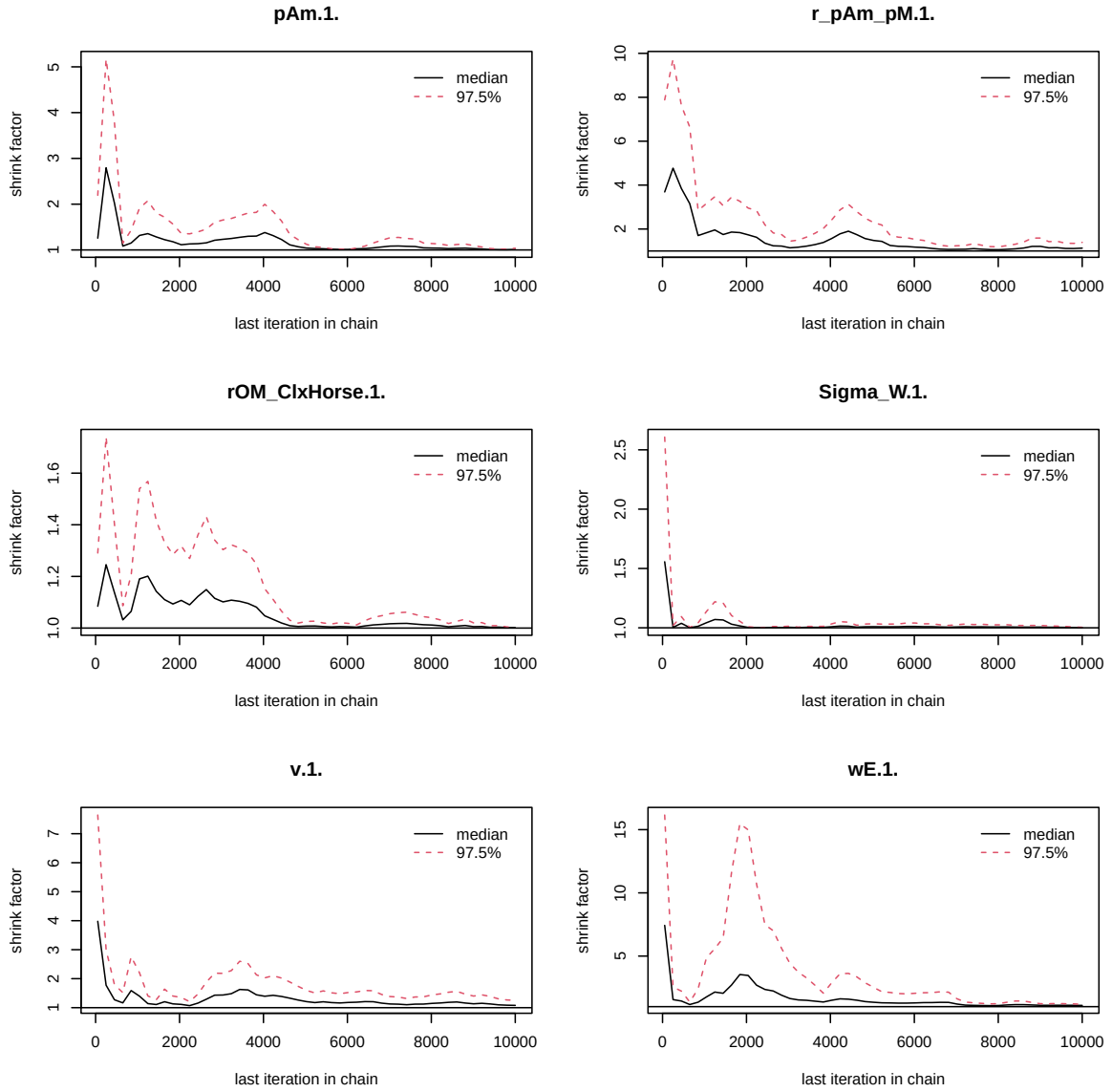


Figure 4: Chains convergence.

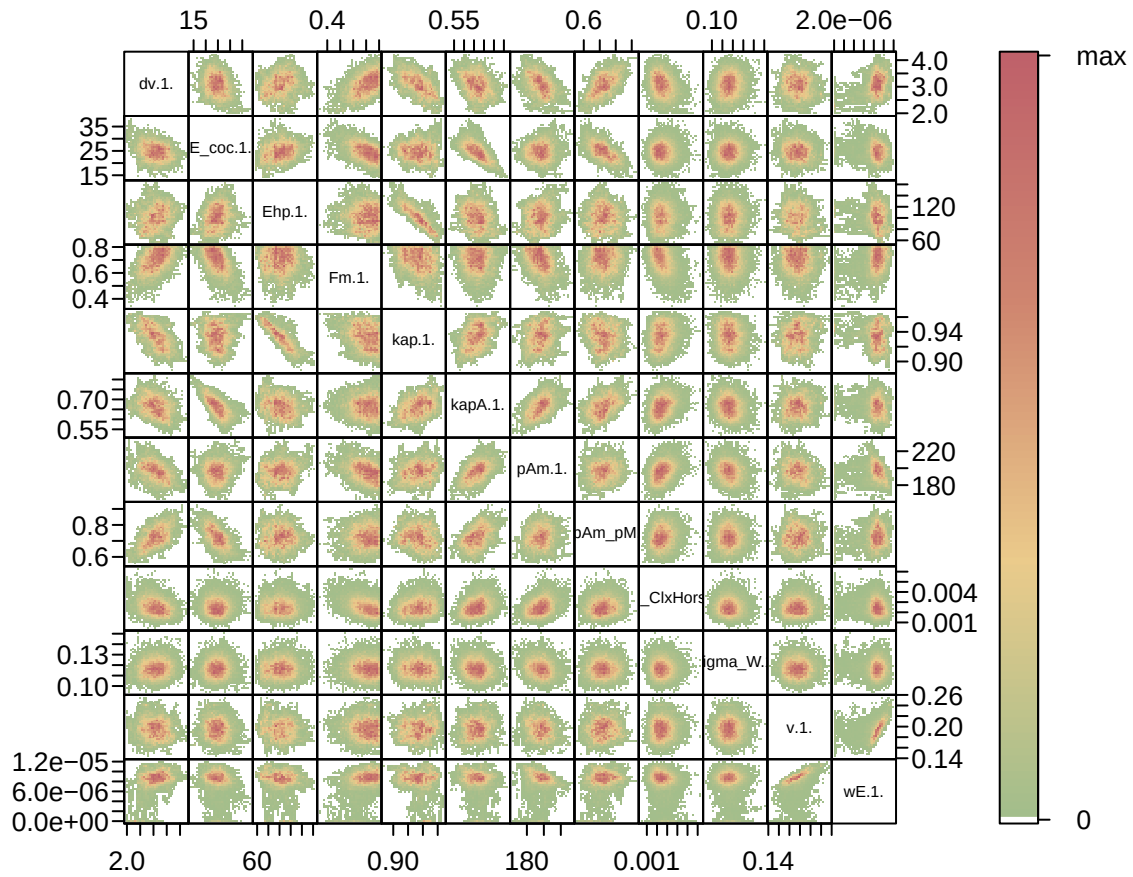


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

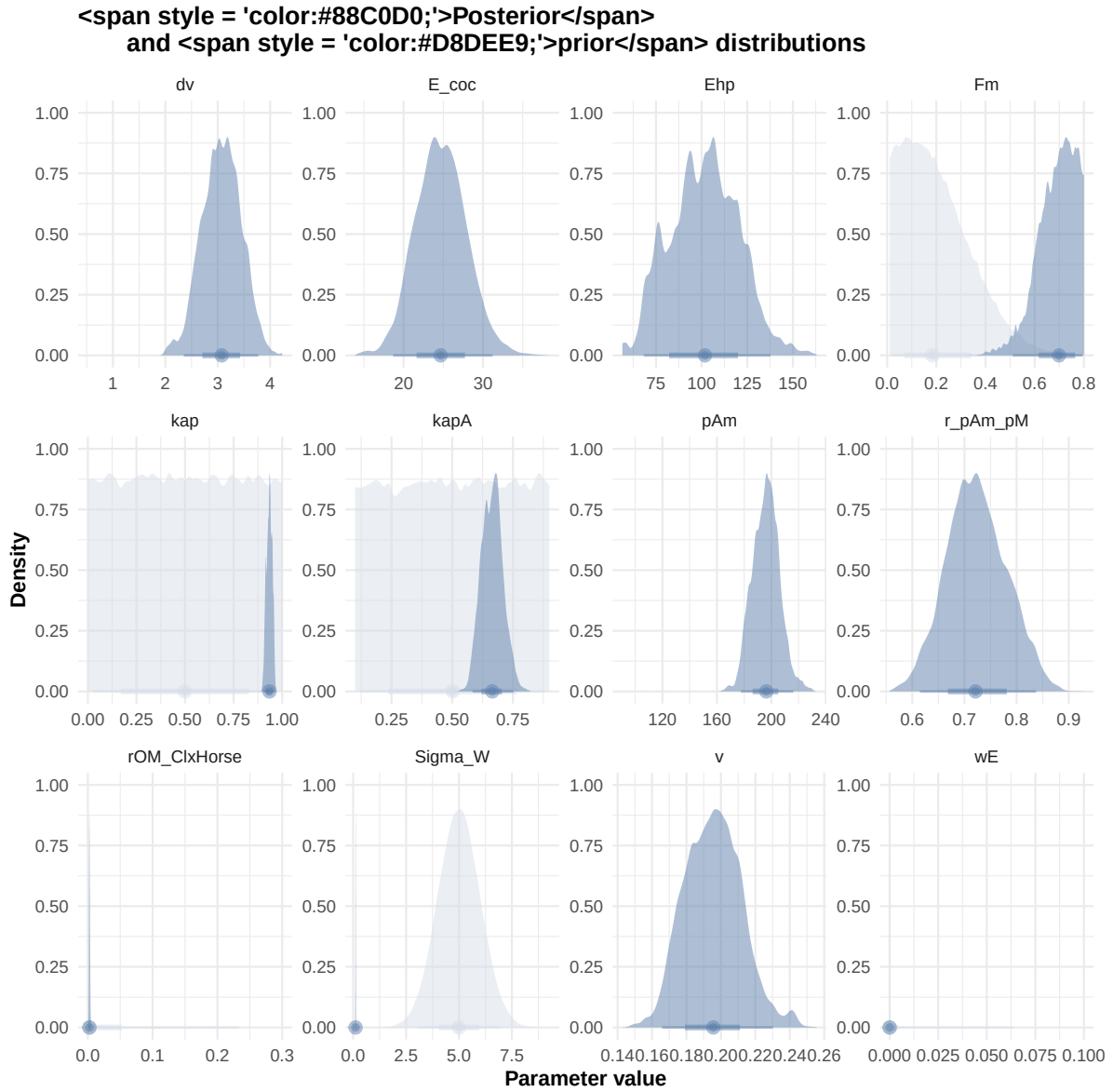


Figure 6: Distributions of the priors and the posteriors.

```

"Bart2020",
#"Gollot2026_lufa"
"Gollot2026_dens_D1",
"Gollot2026_dens_D2",
"Gollot2026_dens_D2b",
"Gollot2026_dens_D5",
"Gollot2026_dens_D7"
)

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\\\.1\\.\\.\\.|\\.\\.\\.1\\.\\.\\.\"", "", param),
    line = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
  Length_struct,
  Reproduction,
  Energy,
  Organic_matter,"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_print)
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr

# ./mcsim.DEBcali DEB_sim_full.in

#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),

```

```

    color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
              Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
  )
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

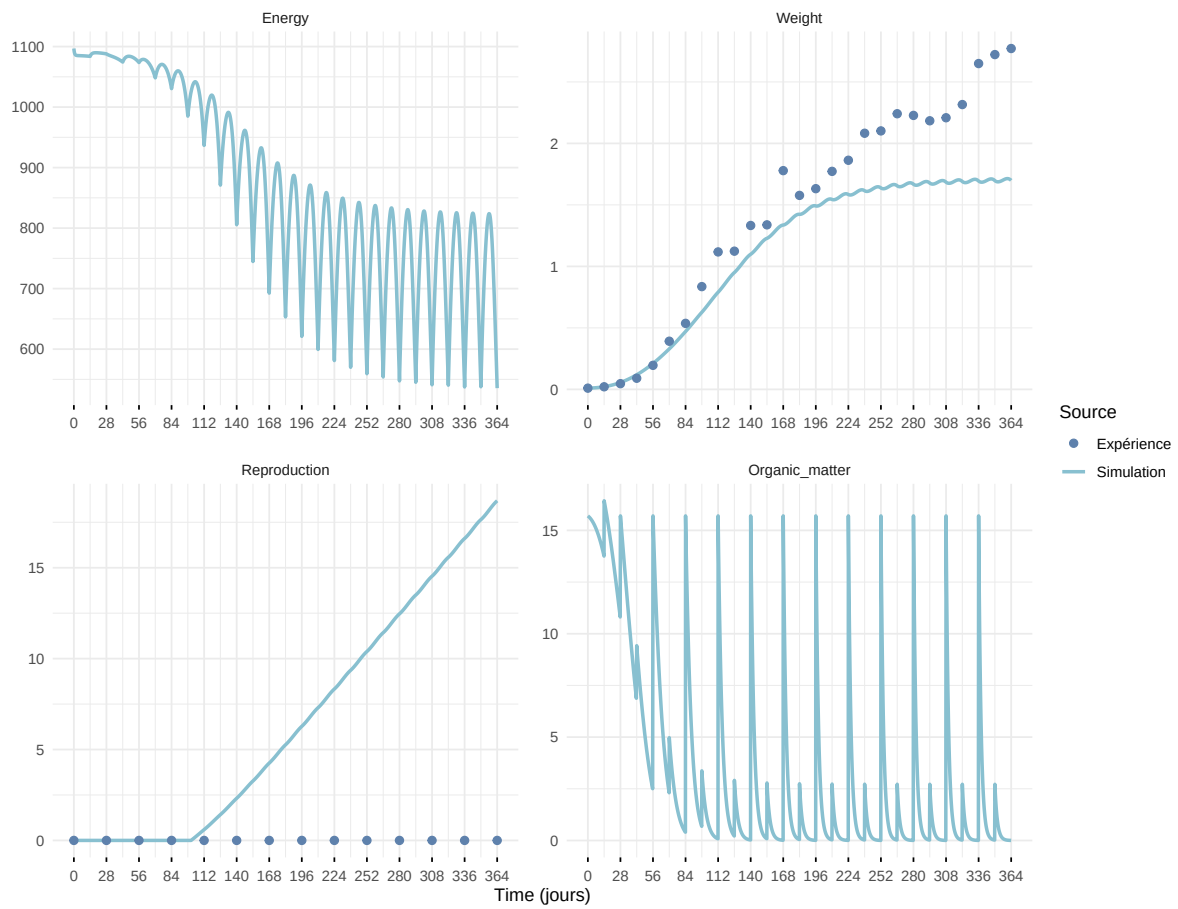
df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
)

```

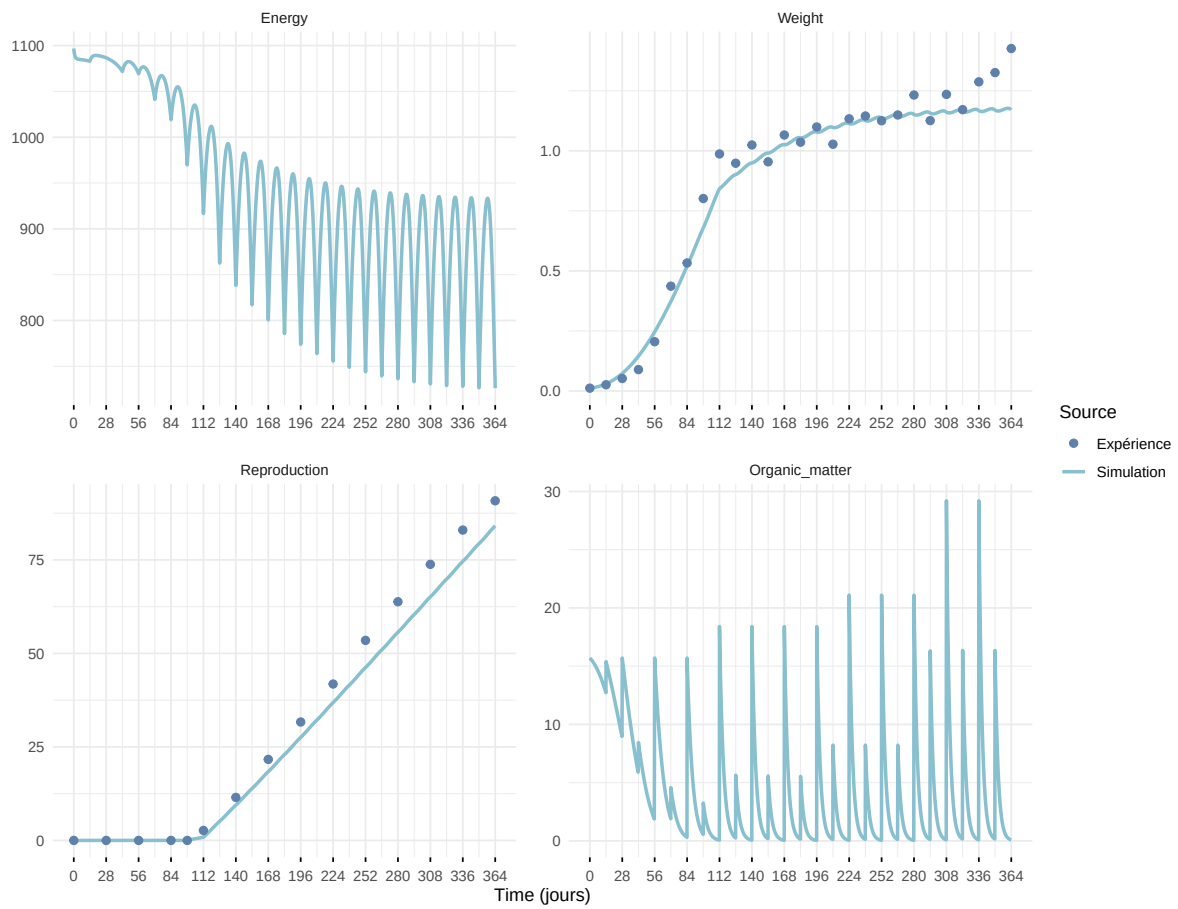


Figure 7: Simulations Weight and Reproduction.

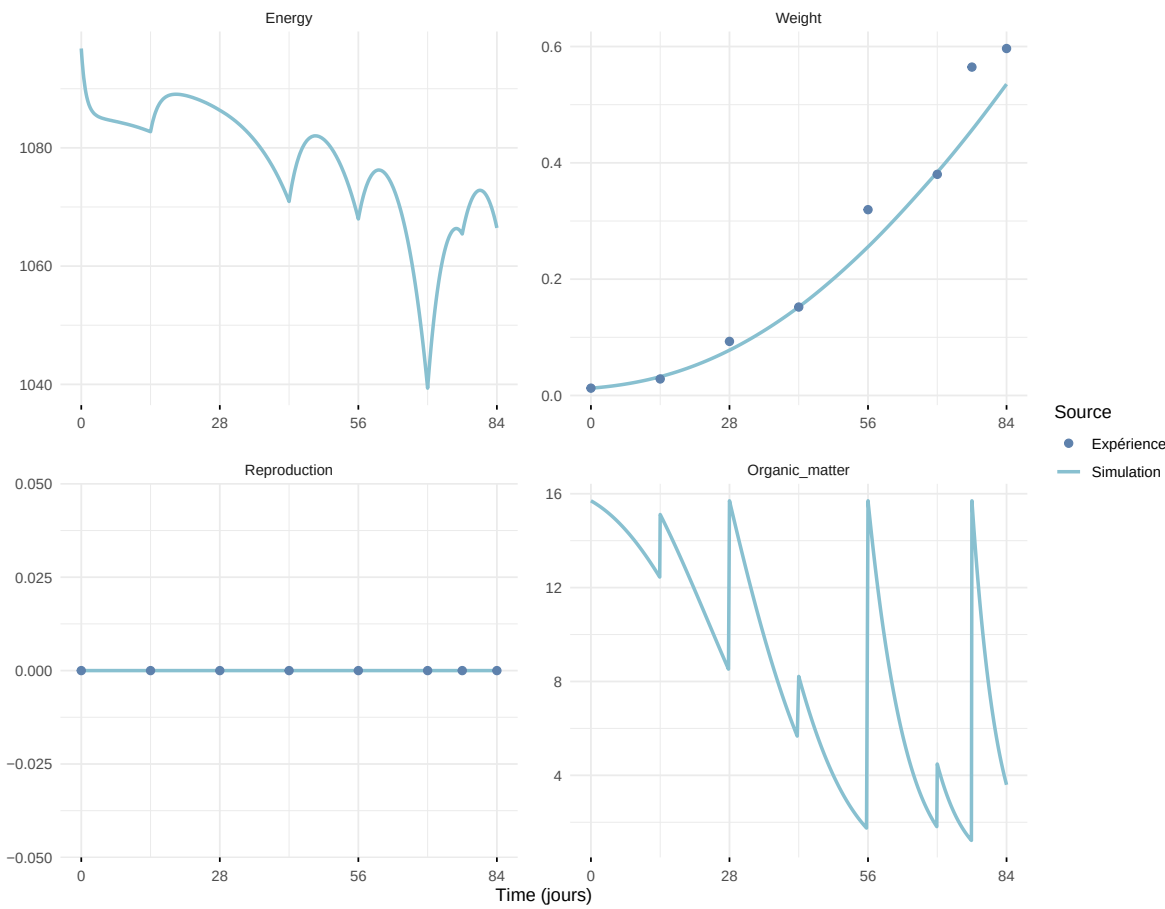
Expérience : Bart2019_XP1_1



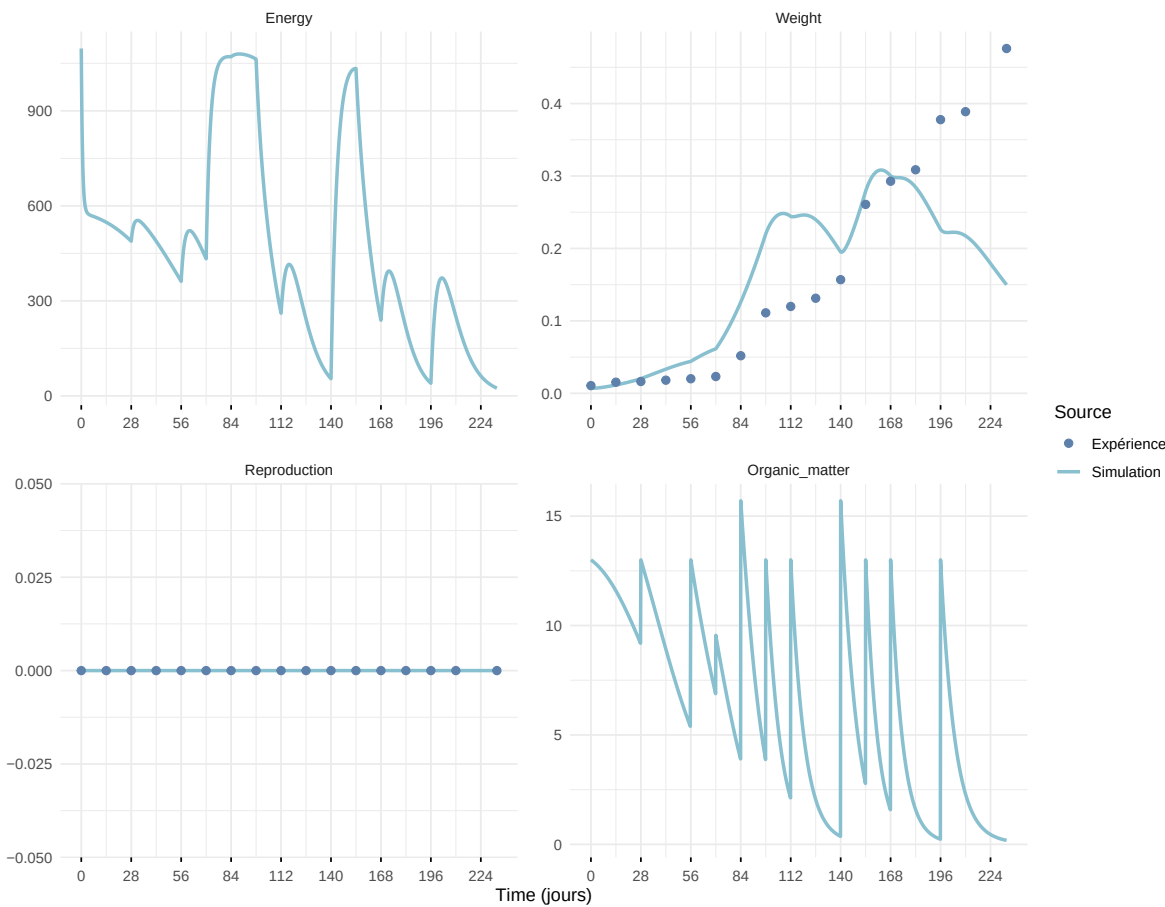
Expérience : Bart2019_XP1_2



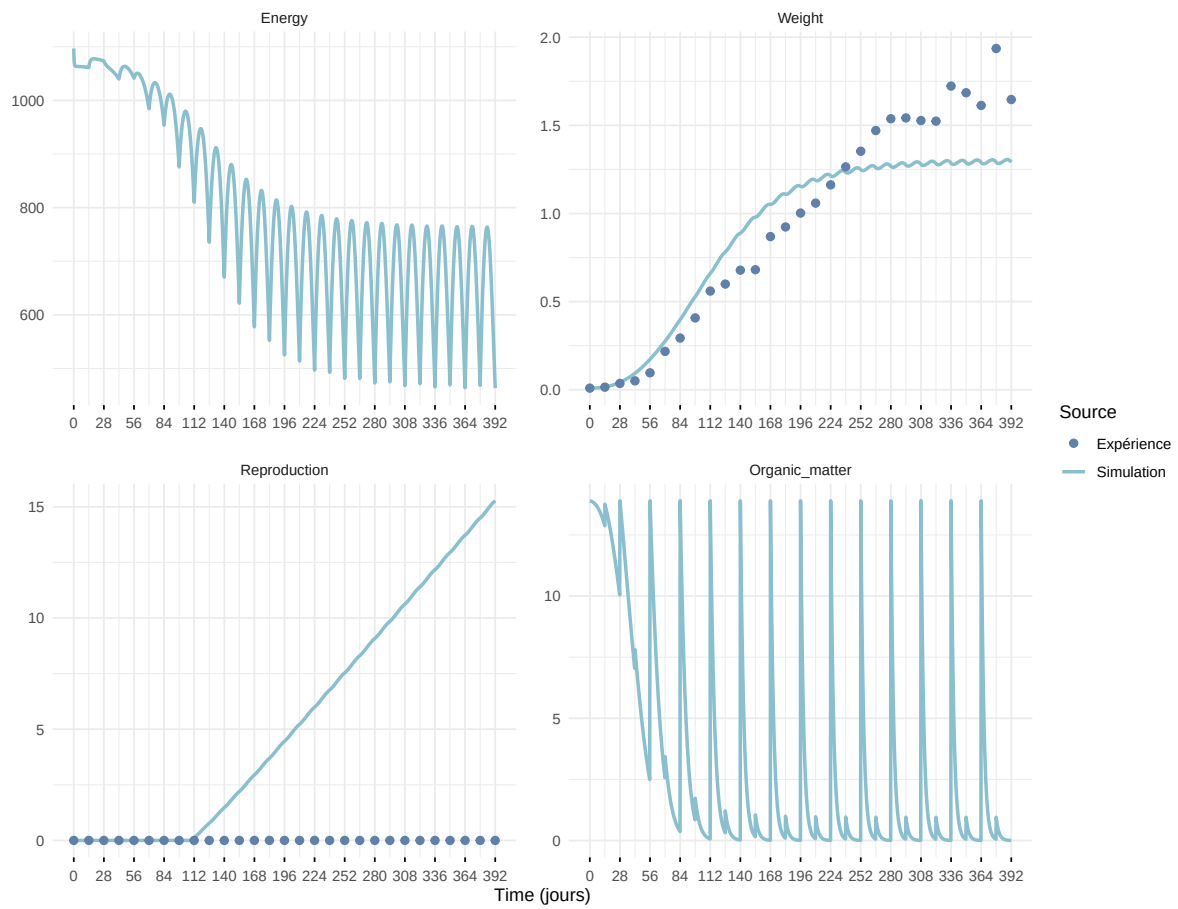
Expérience : Bart2019_XP2



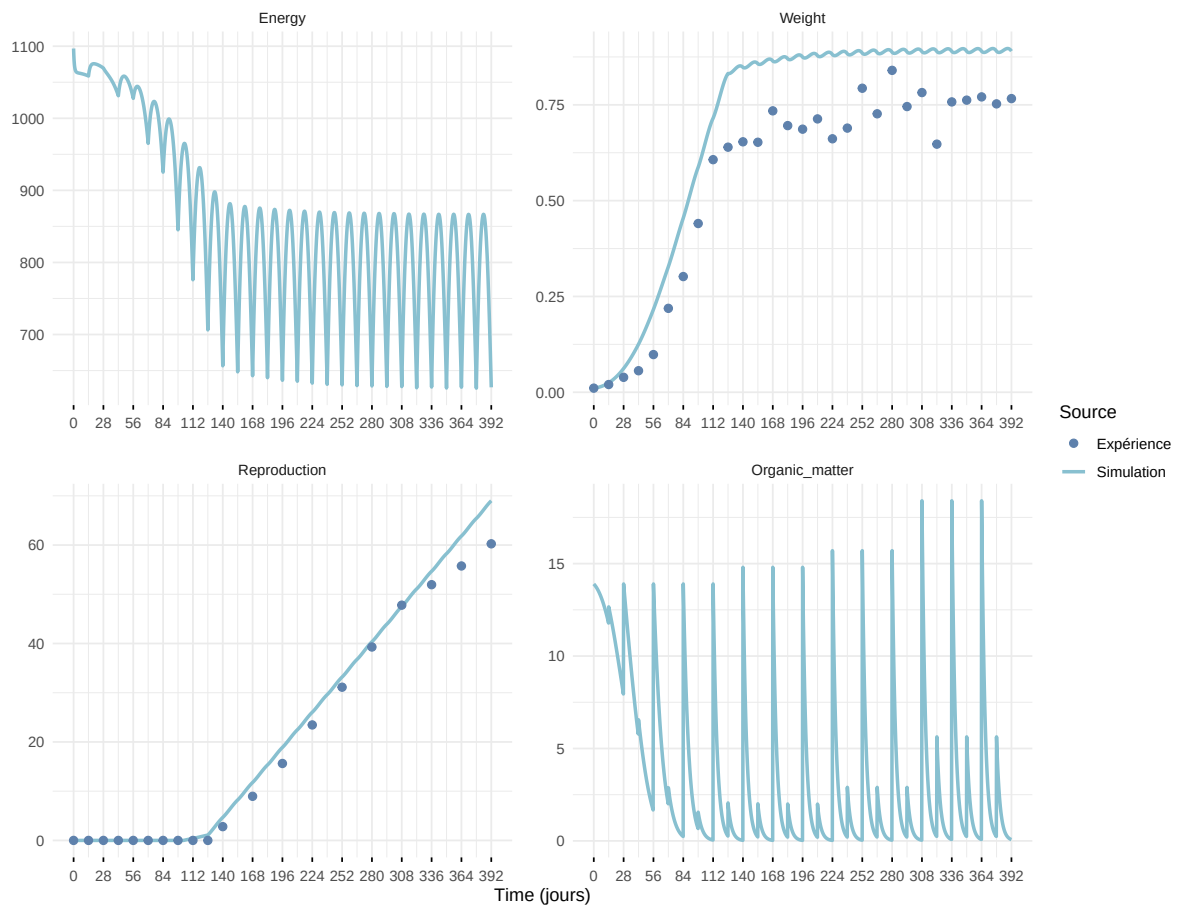
Expérience : Bart2019_XP3



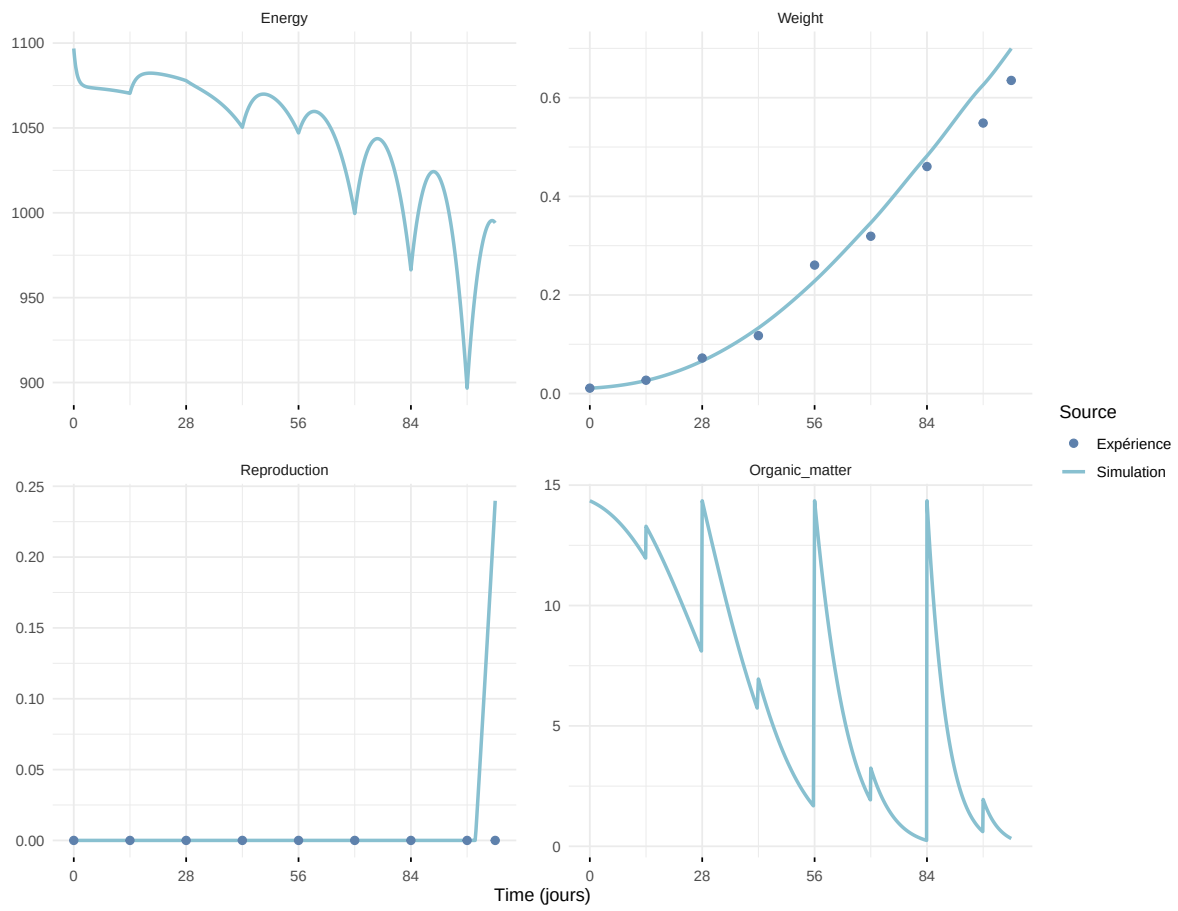
Expérience : Bart2019_XP4_1



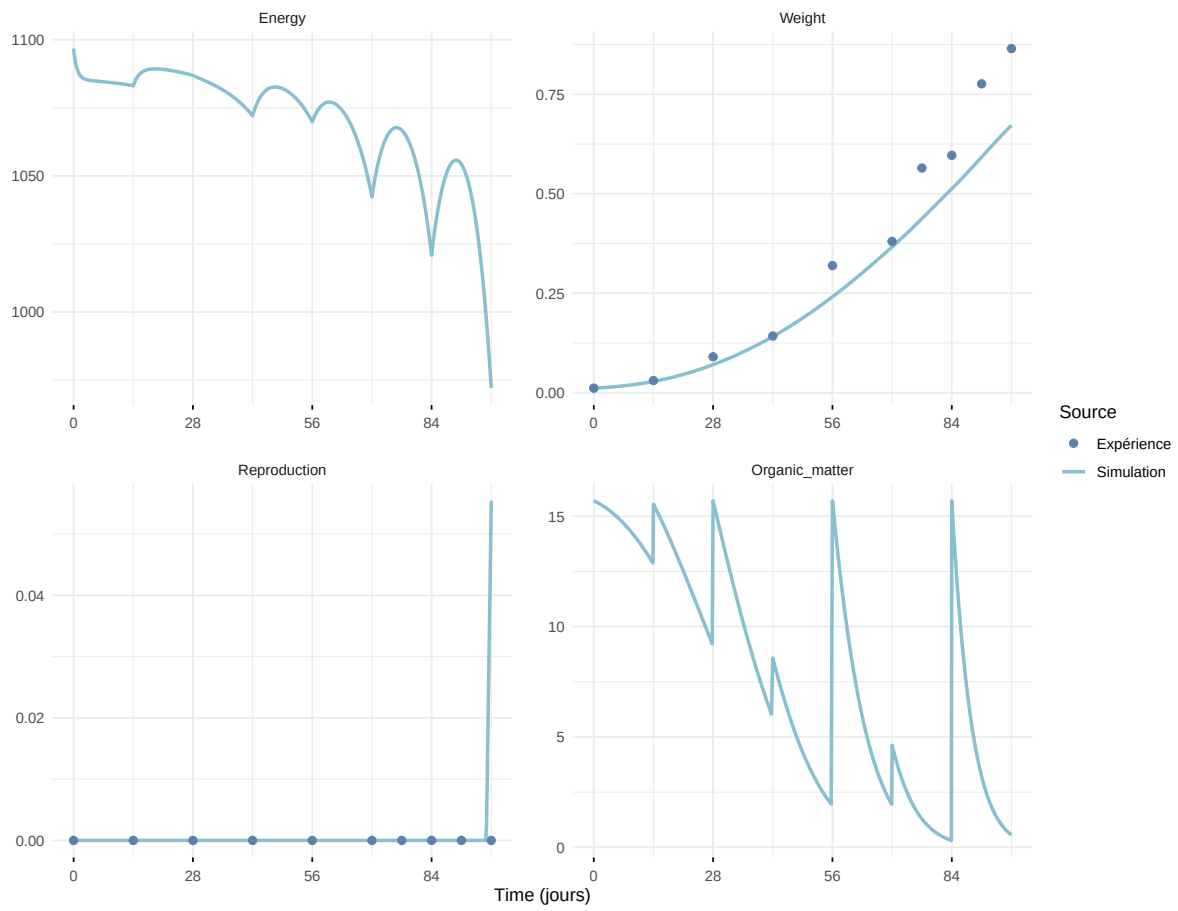
Expérience : Bart2019_XP4_2



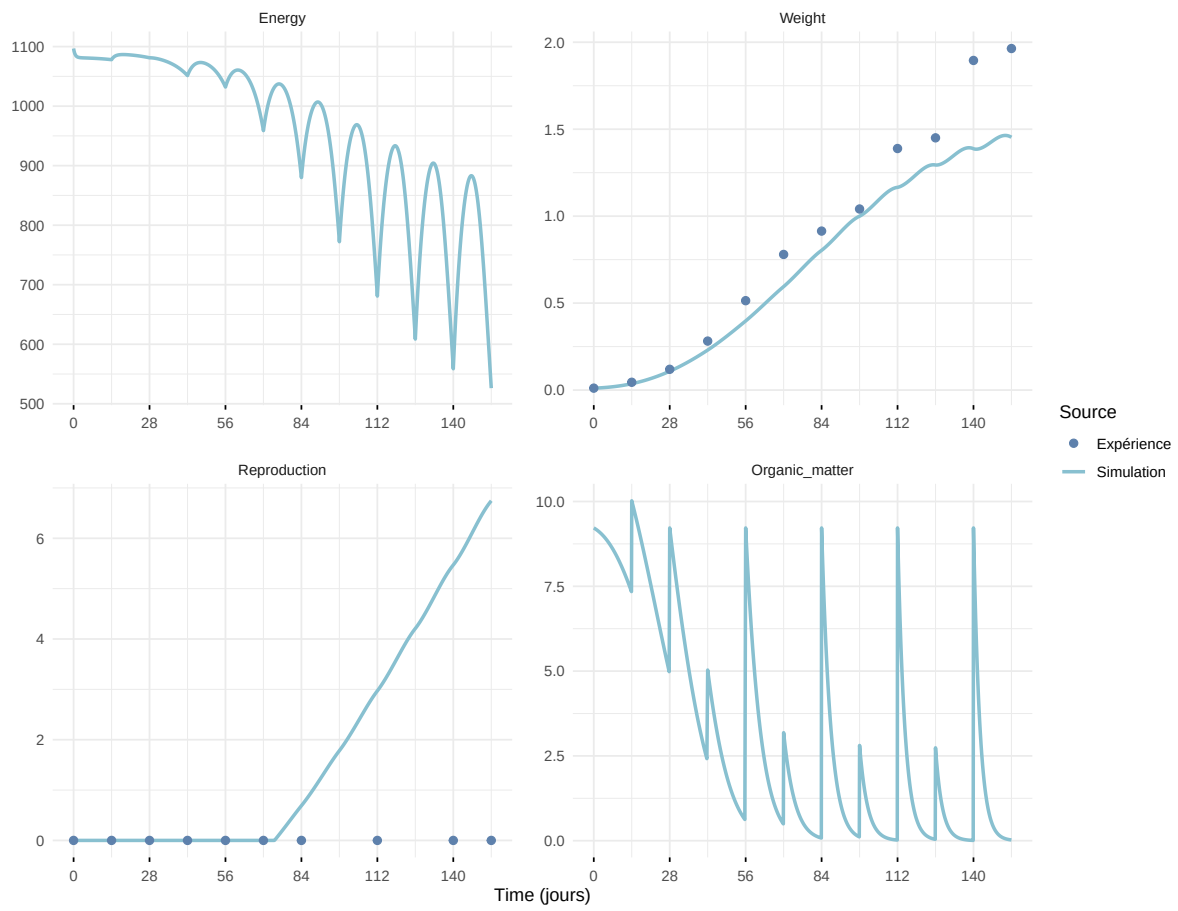
Expérience : Bart2019_XP5



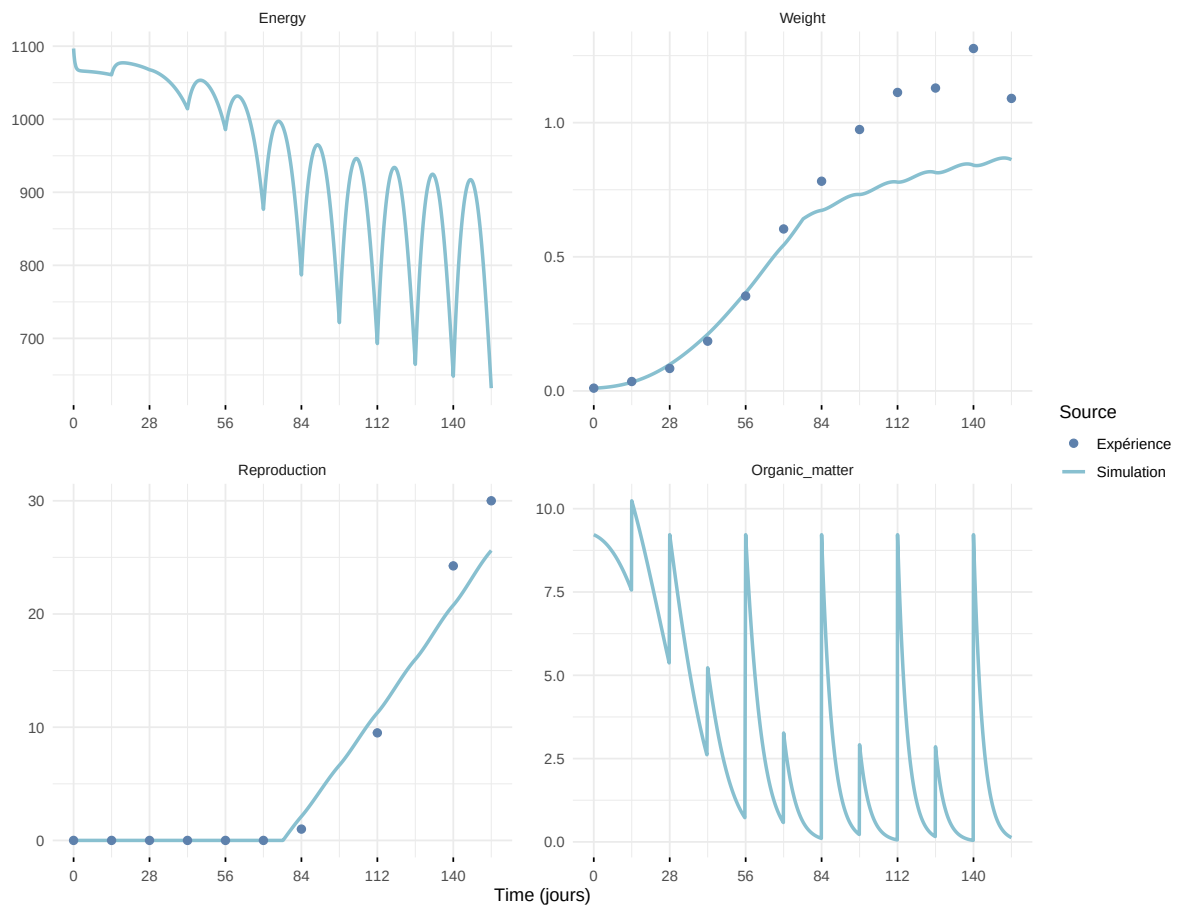
Expérience : Bart2020



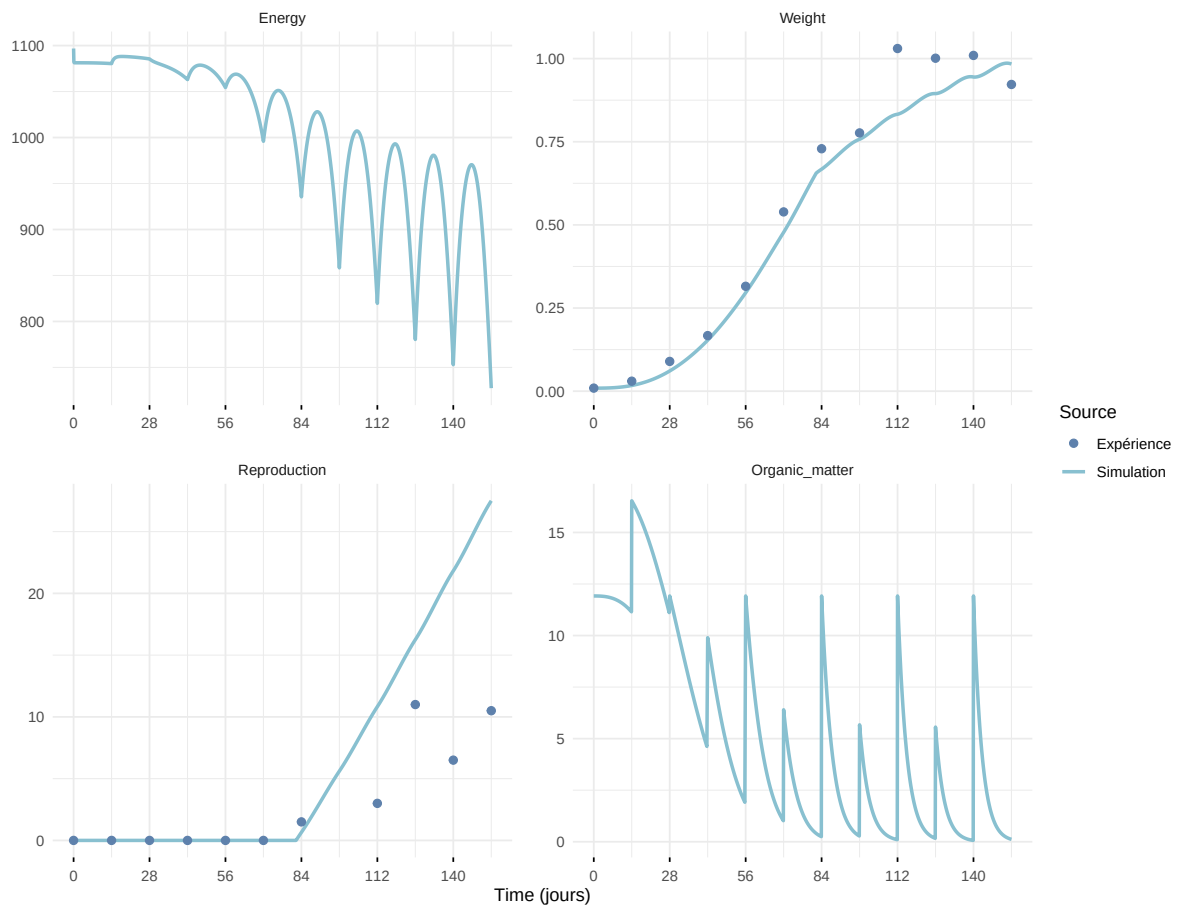
Expérience : Gollot2026_dens_D1



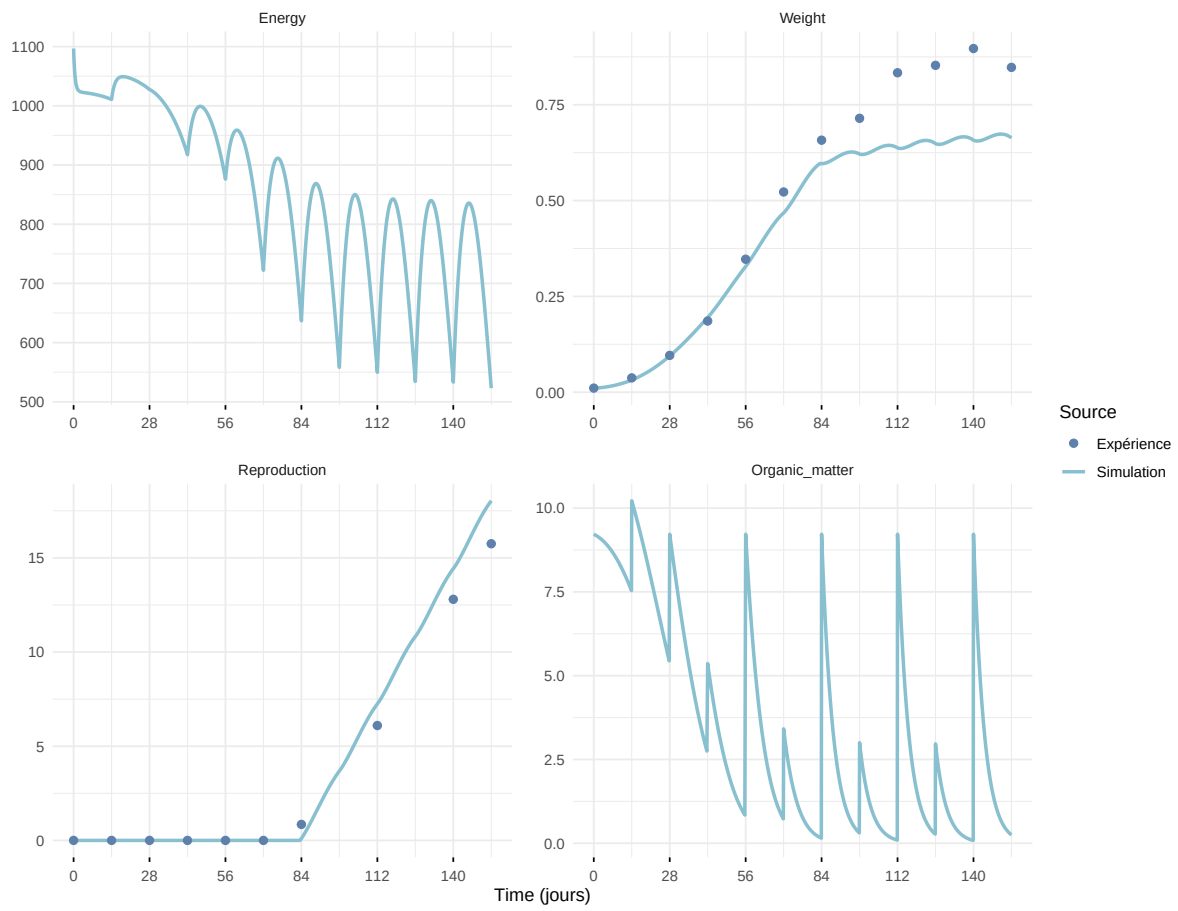
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

